

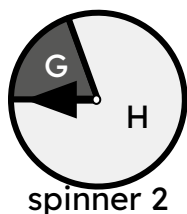
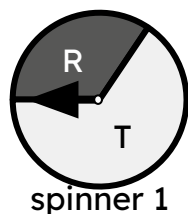


Calculating theoretical probabilities from probability trees (two events)

Task A: Probability of an outcome from a two-stage trial

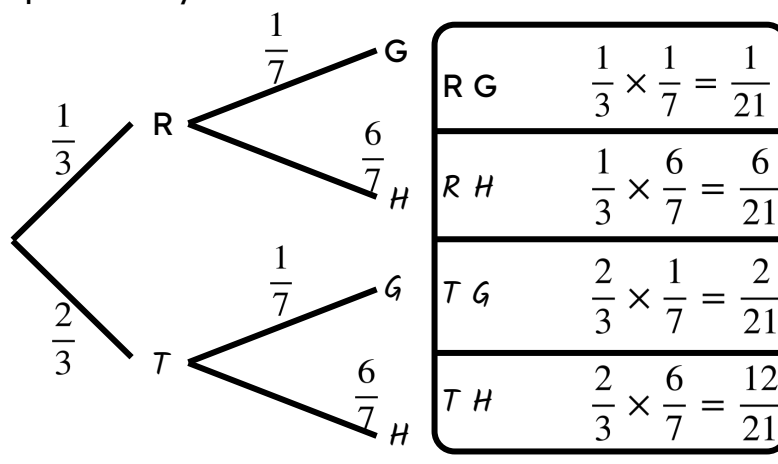
1) Spinner 1 is spun several times.
The spinner landed on R $\frac{1}{3}$ of the time.

Spinner 2 is spun several times.
The spinner landed on G $\frac{1}{7}$ of the time.



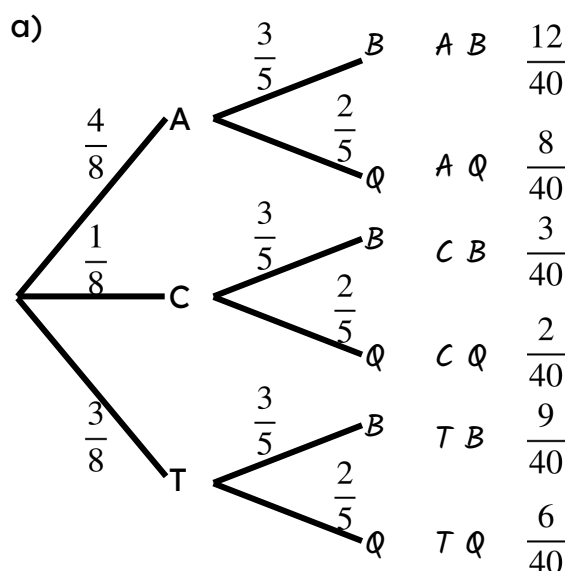
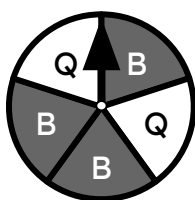
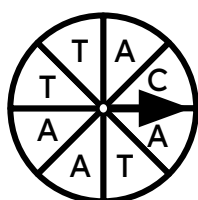
Trial: spinning spinner 1 and spinner 2 once each.

Complete the probability tree to calculate the probability of each outcome from this trial.



2) Trial: spinner 1 and spinner 2 are spun once each.

This probability tree currently shows all the outcomes on spinner 1.

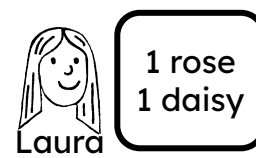
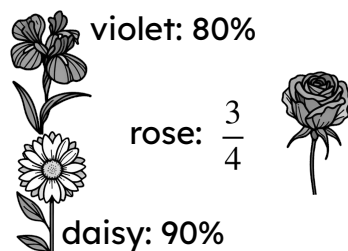


b) What is the probability of the outcome C Q happening? $\frac{2}{40}$

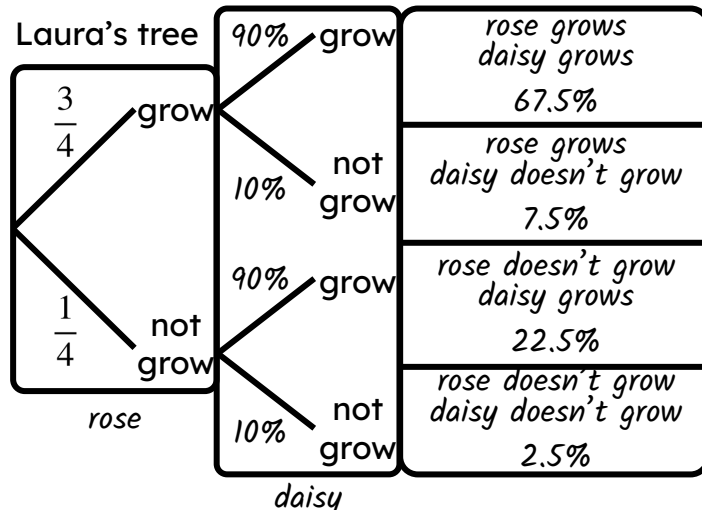
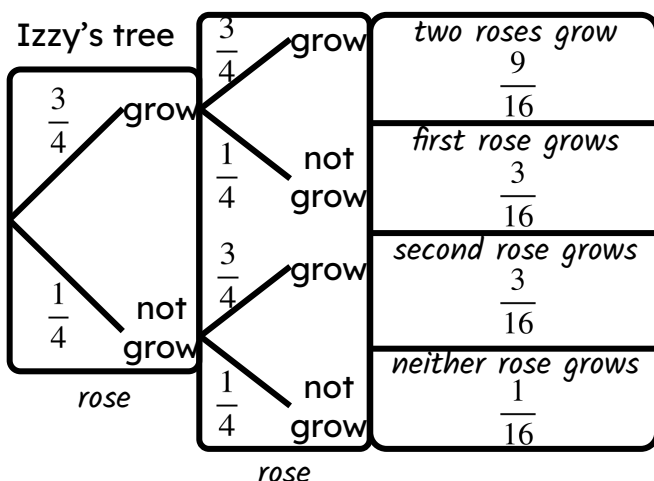
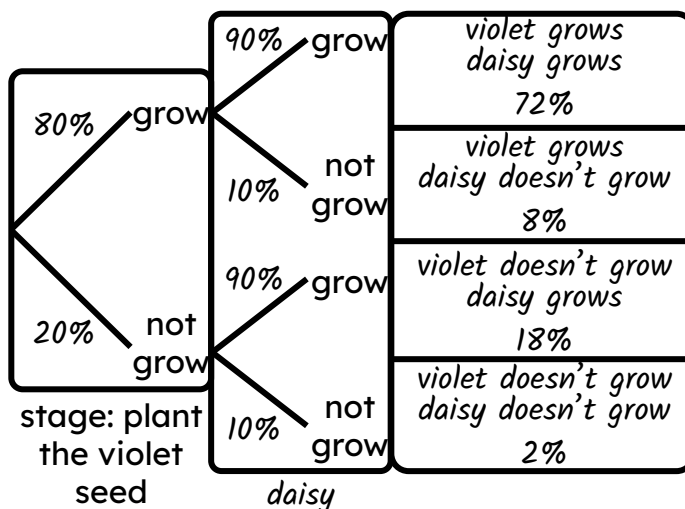


3) Lucas, Izzy, and Laura each take two flower seeds from a packet to grow.

The probability of each type of flower growing is written on the packet.



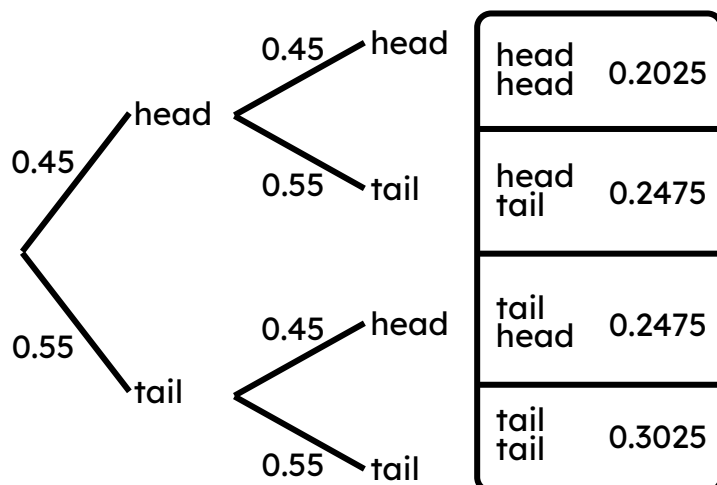
Lucas' tree



Task B: Probability of an event from a two-stage trial

1) A coin is flipped several times. On 45% (or 0.45 as a decimal) of trials, the coin lands on heads.

Trial: flipping this coin twice.



a) What is the probability of the outcome "two heads"?

$$0.2025$$

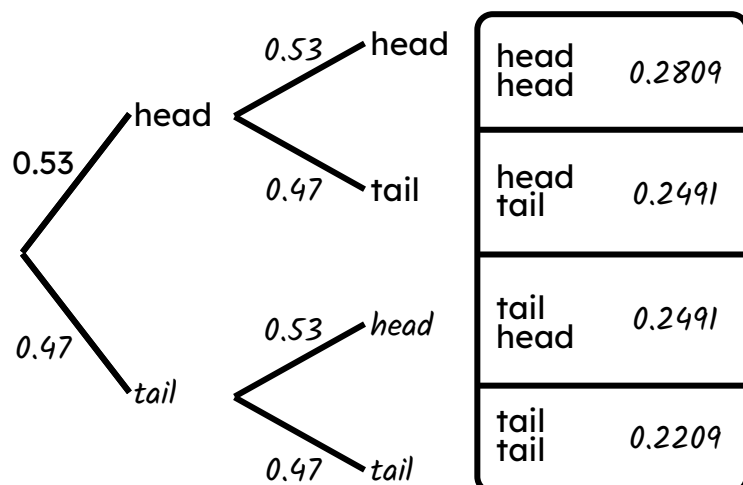
b) What is the probability of the event that at least one coin lands on a heads?

$$0.2025 + 2 \times 0.2475 = 0.6975$$

c) What is the probability of the event that exactly one coin lands on a tails?

$$0.2475 + 0.2475 = 0.495$$

2) A different coin is flipped several times. On 53% (or 0.53 as a decimal) of trials, the coin lands on heads.
Trial: flipping this coin twice.



Event 1: landing on heads twice.
Event 2: landing on exactly one head.

Event 1: 0.2809

Event 2:

$$0.2491 + 0.2491 = 0.4982$$

Event 2 is more likely, by 0.2173 (or 21.73%).

3) Lucas is playing a video game, and needs to scare away a cowboy using either his dragon or unicorn.

The probability of a character hitting their opponent is shown by their “accuracy” stat.

In order for Lucas to scare away the cowboy, either Lucas’ character has to hit, or the cowboy has to miss (or both).

By constructing two probability trees, find out whether the dragon or unicorn is more likely to scare away the cowboy.



dragon
accuracy: 80%



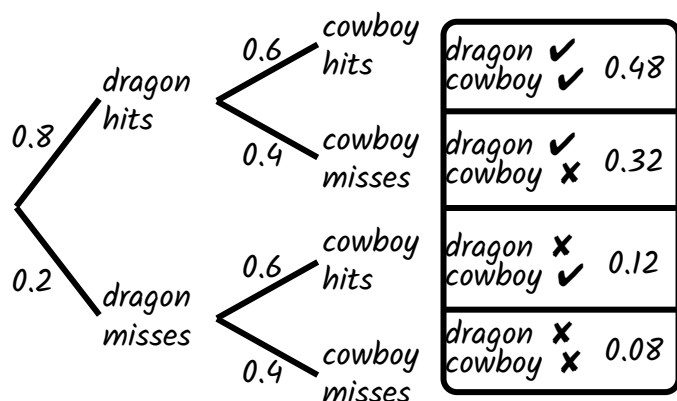
cowboy
accuracy: 60%



unicorn
accuracy: 60%

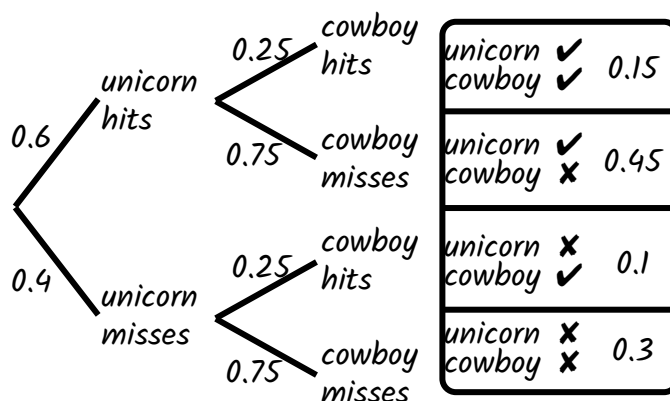


cowboy
accuracy: 25%



The probability that the dragon hits or the cowboy misses is:

$$0.48 + 0.32 + 0.08 = 0.88$$



The probability that the unicorn hits or the cowboy misses is:

$$0.15 + 0.45 + 0.3 = 0.9$$

Lucas should use the unicorn to scare away the cowboy.

